

Experimental investigation on vortex-induced vibrations of a single-box girder with extended cantilevers and its mitigation strategies

Zhen Wang^a, Xuhui He^{a,b}, Haiquan Jing^{a,b}, Lingfei Tan^{a,c}, Naeem Muhammad^a,
Lulu Liu^a, Lei Yan^{a,b,*}

^a School of Civil Engineering, Central South University, Changsha 410075, China

^b Hunan Provincial Key Laboratory for Disaster Prevention and Mitigation of Rail Transit Engineering Structures, Changsha 410075, China

^c China Railway Fourth Survey and Design Institute Group Co., Ltd., Wuhan 430056, China

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ABSTRACT

This study experimentally investigates the vortex-induced vibrations (VIVs) of a single-box girder with long cantilevers, focusing on the impact of handrail modifications on mitigating VIV amplitudes. The results highlight that the VIVs of the bridge occur in both vertical bending and torsion, with the VIV peaks exceeding the permissible values in both degrees of freedom. The analysis reveals that both the outer handrails and maintenance rails influence the VIV performance, with the outer handrails playing a dominant role. Smoke flow visualization provides significant insights, demonstrating that large amplitude VIVs are closely associated with the curbstone and bottom porosity of the outer handrails. Modifications to the outer handrails, particularly the streamlining of the curbstone's aerodynamic shape and improvements to the bottom porosity, are found to significantly reduce the VIV amplitudes in both vertical bending and torsion. The findings suggest that modifying the outer handrails could be an effective approach to controlling VIVs in similar bridge structures.

1. Introduction

Wind-induced vibrations are a critical consideration in the design of long-span bridges and primarily include buffeting, flutter, and vortex-induced vibration (VIV) [1]. Buffeting is a random response to fluctuating winds over a wide range of wind speeds, which can increase fatigue demands and raise serviceability concerns throughout the bridge's service life [2,3]. Flutter is a self-excited aeroelastic instability arising from the coupled interaction between aerodynamic forces and structural motion. If unchecked, it can rapidly grow to large amplitudes, posing a serious threat to structural safety [4,5]. Consequently, accurately predicting the onset of flutter remains a central focus in the wind-resistant design of long-span bridges [6–11]. VIV is an amplitude-limited vibration caused by periodic vortex shedding at specific wind speeds. Although VIV generally occurs at lower wind speeds than flutter and does not directly cause structural failure, it can compromise ride comfort, accelerate fatigue damage, and, in severe cases, necessitate temporary operational restrictions [12–16].

Box girder bridge decks are widely used as the main structural system in long-span bridges, a choice motivated by their economical merits and

aerodynamic performance. The aerodynamics of box girder bridge decks are characterized by lower wind loading and higher critical flutter speeds. However, quite often they may suffer vortex-induced vibration (VIV) at low wind speeds due to vortex shedding phenomena [17,18]. This fact was highlighted in the last decade by some representative VIV cases of long-span bridges, for example, the Storebælt suspension bridge [12], Osterøy suspension bridge [13], Humen bridge [14–16], etc.

Controlling VIV in box girder bridges has attracted the attention of several researchers. Passive control techniques have been widely studied and applied in engineering practice because of their reliability, simplicity and cost effectiveness. Simple aerodynamic attachments to the leading and trailing edges of bridge decks such as guide vanes [12, 19–22], wind fairing [23–26] and flaps [27,28] have been used to stabilize VIV by optimizing the geometric shape or interfering the surrounding flow field. While aerodynamic attachments constitute an efficient way of mitigating VIV of box girder bridges, their disadvantages reside in the fact that the cost effectiveness is lost due to high maintenance efforts and construction costs [13,29].

To avoid high maintenance efforts and construction costs of aerodynamic attachments, the modification and relocation of bridge

* Corresponding author at: School of Civil Engineering, Central South University, Changsha 410075, China.

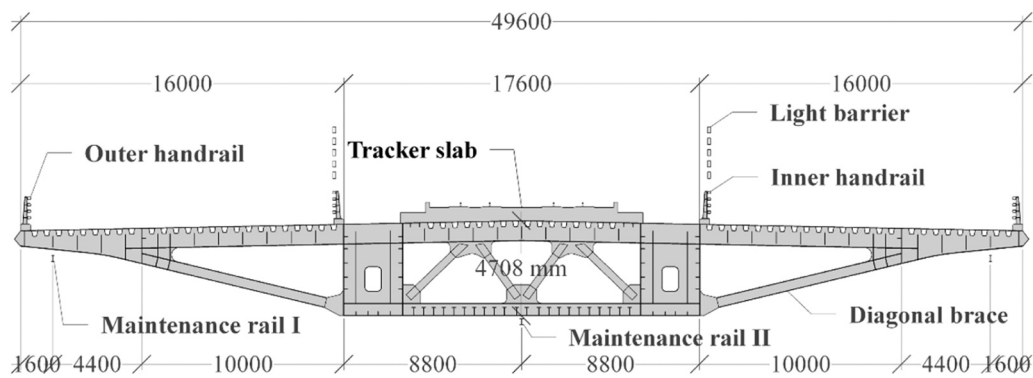
E-mail address: leiyan@csu.edu.cn (L. Yan).

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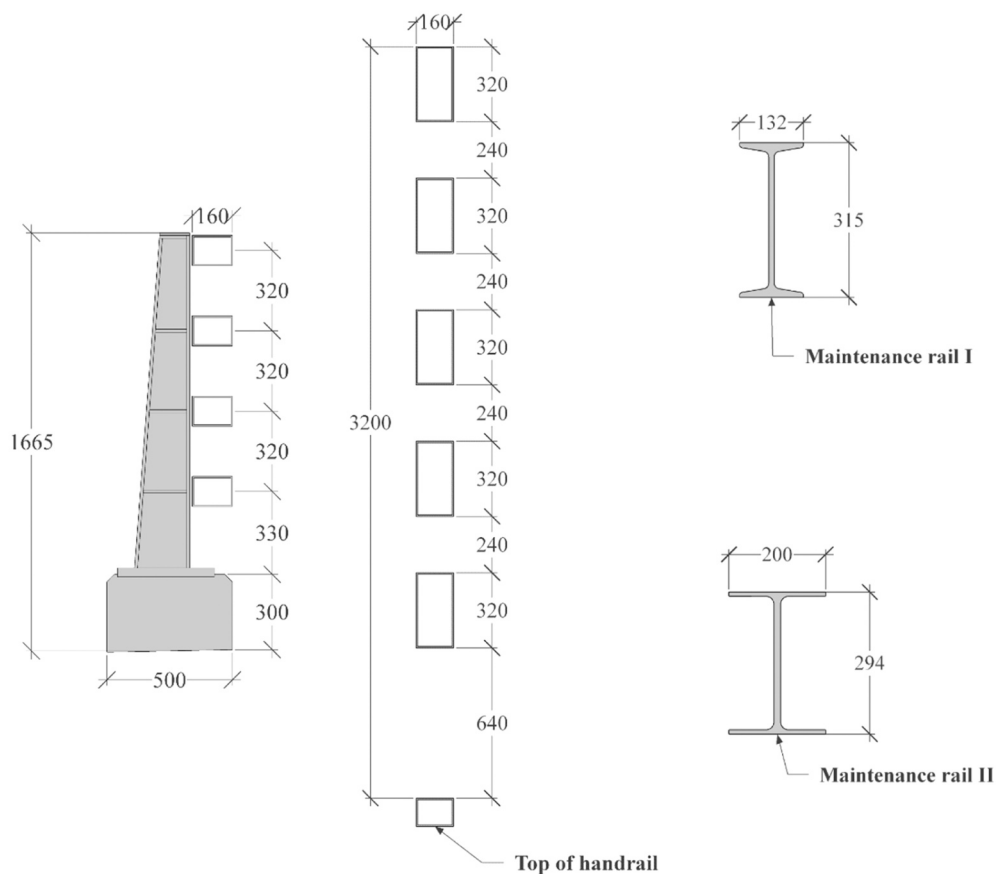
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(a) Cross section



(b) Handrail

(c) Light barrier

(d) Maintenance rail

Fig. 1. Cross section of a highway-railway same-level box girder bridge (unit: mm).

appurtenances (handrails, wind barriers and maintenance rails) have drawn a lot of research interests. The mechanism of suppressing VIV through bridge appurtenances is similar to the mechanism used in aerodynamic attachments, both involve optimizing the shape to improve aerodynamic performance and disrupting the surrounding flow field [30].

Since handrails and wind barriers serve as primary bridge appurtenances with considerable influence on the VIV response, numerous studies concerning handrails and wind barriers have been conducted. Nagao et al. [31] showed that optimizing the size and position of the handrail top horizontal bar reduces the VIV amplitude. Matsumoto et al. [32] experimentally found that the handrail porosity ratio significantly

influenced the maximum VIV amplitude. Haque et al. [24] and Wang et al. [25] studied the effect of the handrails porosity ratio on VIV performance in long-span bridges. They concluded that high-porosity handrails enhance aerodynamic stability by preventing separation bubbles and narrowing wake size, ultimately reducing the VIV amplitude. This improvement is attributed to the flow dynamics on the upper surface of the girder, leading to the elimination of the upper girder vortex. Li et al. [33] successfully managed to suppress VIV in a wide streamline box girder bridge by increasing the porosity ratio of handrails. Yan et al. [34] systematically investigated the influence of the detailed configuration of handrails on VIV performance in a box girder bridge. Their research confirmed that the VIV amplitude exhibited a

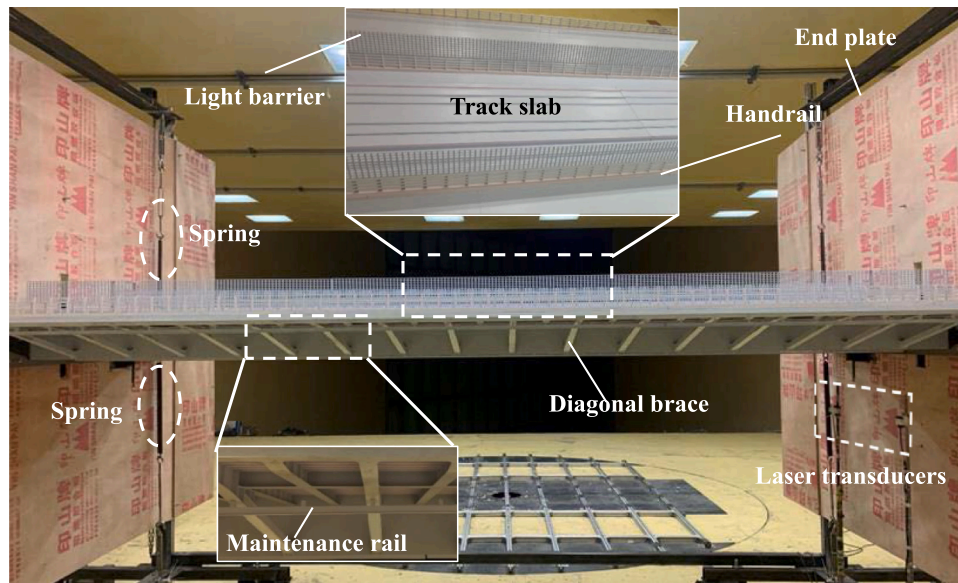


Fig. 2. Installation of a road-rail dual-use box girder in the CSU-WT1.

Table 1
Dynamic parameters of the section model.

Parameter	Full-Scale bridge	Scale ratio	Section model
Width B (m)	49.600	1:25	1.984
Length L (m)	-	1:25	4.000
Height H (m)	4.708	1:25	0.180
Mass m (kg/m)	64029	1:25 ²	101.250
Mass moment of inertia J (kg•m ² /m)	8698420	1:25 ⁴	22.268
Vertical frequency f_h (Hz)	0.286	8.389:1	2.417
Torsional frequency f_t (Hz)	0.550	8.389:1	4.699
Vertical damping ratio ξ_h (%)	-	-	0.27
Torsional damping ratio ξ_t (%)	-	-	0.30

decrease as the porosity ratio of handrails increased. Furthermore, they found that there were distinctive aerodynamic differences between grid-like and non-grid handrails. In the case of non-grid handrails, increasing the number and side ratio of middle horizontal bars resulted in a reduction of the VIV amplitude. Conversely, for grid-like handrails, an increase in the number of vertical bars adversely affected VIV, while the number of horizontal bars showed no impact on VIV. Sun et al. [35] conducted wind tunnel tests to examine the effects of bridge appurtenance, such as handrails, wind fairings, crash barriers, and maintenance

rails, on the VIV performance of a streamlined closed-box girder. The results indicated that optimizing the position, installation angle, and height of the handrails can effectively improve VIV performance.

Wind barriers exert a similar influence on the VIV performance of bridges as observed with handrails. Kozmar et al. [36,37] experimentally found that wind barriers have the capability to reduce wind velocity on a bridge girder, and an increase in barrier porosity correlates with a reduction in vorticity. Buljac et al. [38,39] validated that wind barriers exert a significant impact on aerodynamic forces, reducing the torsional flutter stability of bridge-deck sections. Liu et al. [40] successfully reduced the VIV amplitude of a twin separated parallel box girder bridge by modifying the height and porosity of wind barriers. However, Wu et al. [41] noticed that the unsteady shear flow separating from the top of the wind barrier exerts forces on the bridge girders, leading to large-amplitude VIVs of a multi-box girder bridge.

Introducing three-dimensional disturbance in the flow field is another idea to control VIV through handrails and wind barriers. Zhan et al. [42] introduced three-dimensional disturbance through a wavy handrail with vertical posts and wavy crossbars and effectively eliminated both vertical and torsional VIV when the wavelength of the wavy handrails is 0.5–1 time of the girder height. Despite the efficacy of wavy handrails in VIV mitigation, practical application in bridge engineering is hindered due to space requirements on the bridge deck. Bai et al. [43] proposed a sealed traffic barrier to introduce three-dimensional

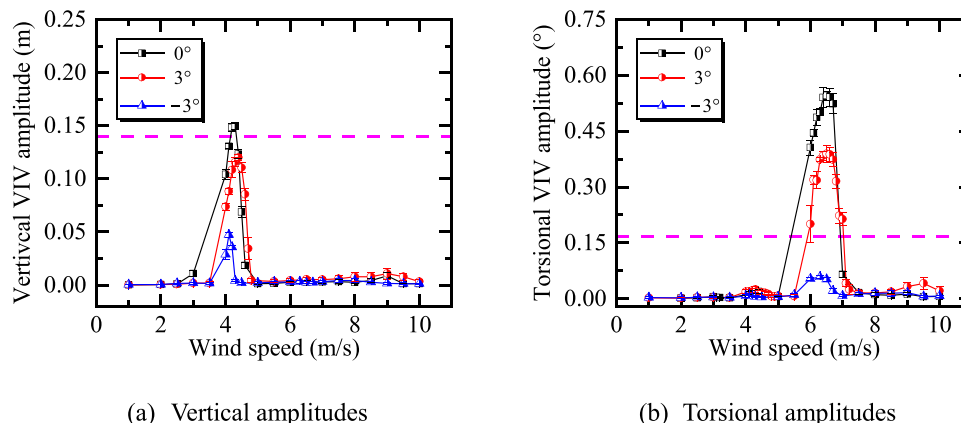


Fig. 3. VIV amplitude for original bridge deck.

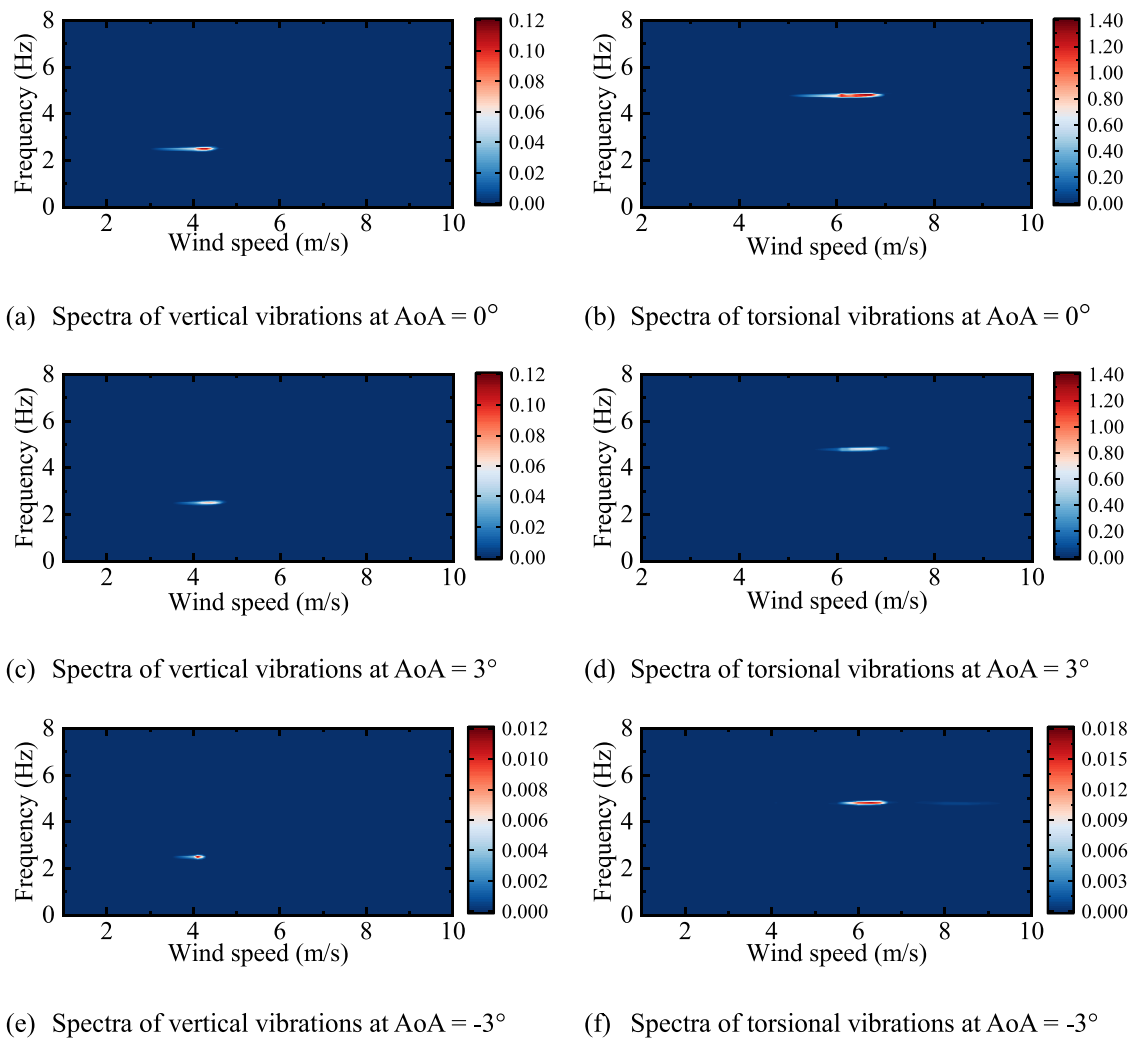


Fig. 4. Vibration spectra for bridge section model.

disturbance. They experimentally showed that the partially sealed traffic barriers had the potential to disturb the periodic vortex shedding on the bridge surface or diminish the energy of the aerodynamic forces associated with this shedding, effectively suppressing the VIV. Employing a similar approach, Liu et al. [44] effectively reduced the VIV amplitude of a twin separated parallel box girder bridge by the partially sealed handrails.

Maintenance rails also serve as a crucial component influencing the aerodynamic configurations of the bridge deck. Laima et al. [45] found that large flow separation generated by maintenance rails resulted in large pressure fluctuations, a wide wake, and a large-scale vortex shedding, reducing the VIV performance. Sun et al. [46] confirmed that the vortex shedding induced by maintenance rails had a significant effect on the VIV performance of a box girder bridge. Li et al. [47] experimentally revealed that their location significantly influences aeroelastic behavior. Moving maintenance rails towards the center or perpendicularly away from the bottom surface reduced the VIV amplitude. In addition, attaching guide vanes to maintenance rails further enhanced aerodynamic performance. However, the studies such as Li et al. [33] and Wu et al. [41] showed that maintenance rails had slight effect on reducing the VIV amplitude.

The above literature review demonstrates that modifying bridge appurtenances holds promise for mitigating VIVs in box girder bridge decks. However, most prior studies have concentrated on either highway or railway box girder bridges individually, with little attention paid to the more complex highway–railway same-level box girder bridges. As

China's infrastructure continues to advance, available bridge sites are becoming increasingly scarce. Consequently, highway–railway same-level bridges are gaining prominence owing to their superior traffic capacity and cost-effectiveness. Due to the co-location of highway and railway components on a single bridge deck, the aerodynamic characteristics of these bridges differ markedly from those of single-purpose box girder bridges. As a result, existing aerodynamic control strategies and findings cannot be directly applied to this bridge type. To date, no systematic investigation has examined the VIV performance or aerodynamic optimization of highway–railway same-level box girder bridges, leaving a critical gap in engineering practice.

In the present study, the VIV performance of a highway–railway same-level bridge, characterized by a single-box girder with extended cantilevers, was experimentally investigated. To enhance VIV performance, the effects of maintenance rails and handrails were systematically analyzed, with handrails showing a significant impact. Smoke flow visualization was employed to reveal the underlying VIV mechanisms and guide aerodynamic optimization of the handrail profiles. Through redesigning the handrails, VIV was effectively controlled, demonstrating a successful aerodynamic mitigation strategy for this bridge type.

2. Section model tests

2.1. Engineering background

The prototype of this study is a four-tower cable-stayed bridge in

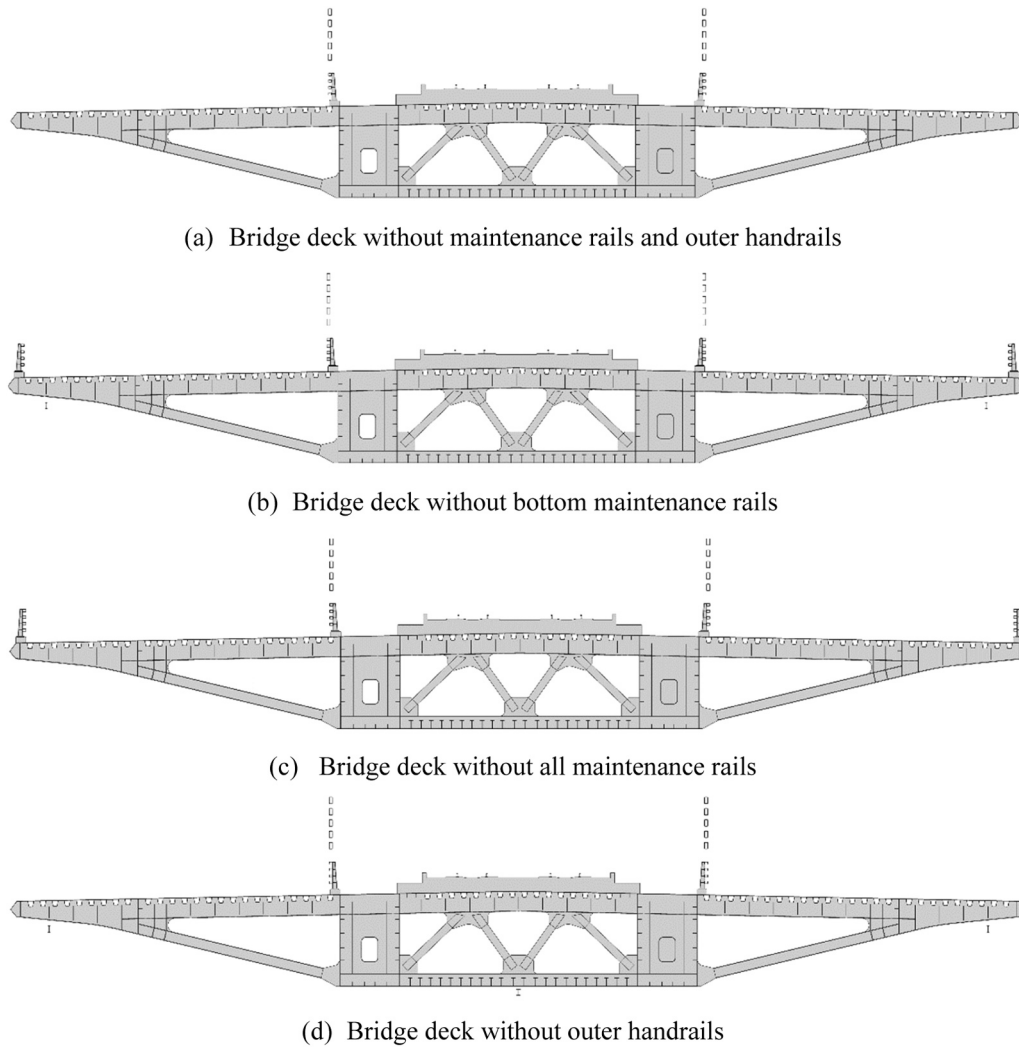


Fig. 5. Schematic of tested bridge decks.

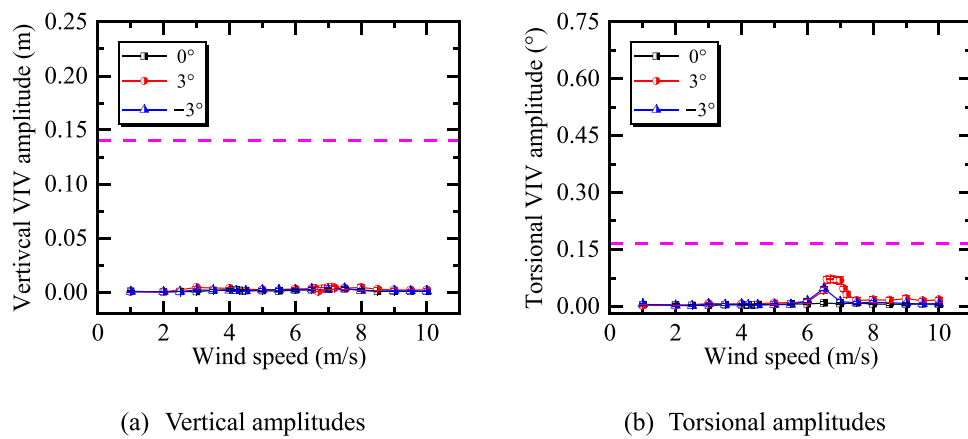


Fig. 6. VIV amplitudes for bridge deck without maintenance rails and outer handrails.

China, which has a layout of $58.5 + 116 + 3 \times 340 + 116 + 58.5$ m. The bridge deck is designed as a highway–railway same-level box girder with a width of 49.6 m because of superior traffic capacity and cost effectiveness. The cross-section of the main girder is designed as a single-box girder with long cantilevers, as shown in Fig. 1. Outside diagonal braces are supplemented at an interval of 6.0 m along the axial direction to

enhance the load-bearing capacity of the cantilever, as shown in Fig. 2. A double track railway with a width of 17.6 m is arranged in the center of the main girder, and a three-lane highway with a width of 16 m is installed on each side. The bridge appurtenances of the highway–railway same-level bridge is more complex compared to the individual highway or railway box girders. The general configuration of the bridge

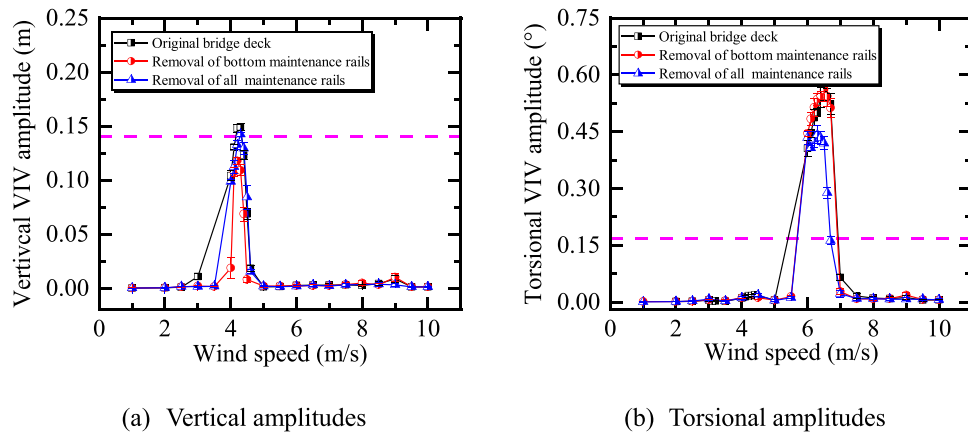


Fig. 7. VIV amplitudes for bridge deck without maintenance rails.

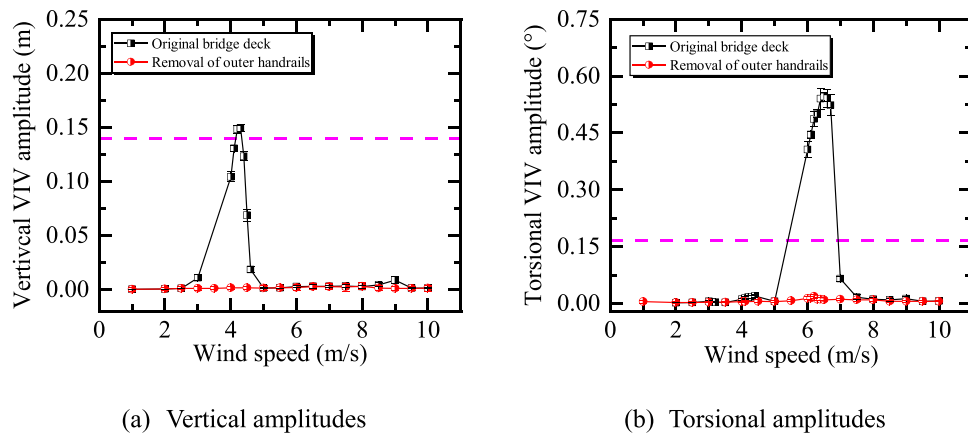


Fig. 8. VIV amplitudes for bridge deck without outer handrails.

deck is shown in Fig. 1, which includes the cross section and bridge appurtenances, such as light barriers, track slabs, handrails, and maintenance rails. Each side of the highway features a pair of handrails, located on either side of the travel lane. The track slabs are positioned at the center of the bridge deck, while the light barriers are strategically placed on the top of inner handrails to prevent light interference. Three maintenance rails are integrated into the bottom of the main girder, with one positioned at the center and the other two symmetrically placed on either side.

2.2. Bridge deck section model

A 1:25 scale model of the bridge deck section was used for testing as shown in Fig. 2, which saved most of the details of the real bridge deck, such as the slight outboard slope, light barriers and track slabs. The dimensions of the section model were 4.0 m length, 1.984 m width, and 0.180 m height. As a result, an aspect ratio (length/width) of approximately 2.02 was achieved, satisfying the wind-resistant design specification [48]. The model shape and the bridge appurtenances were manufactured using plastic materials (PVC). The aluminum frame composed of three longitudinal beams and six cross beams was installed inside the model to ensure the model rigidity.

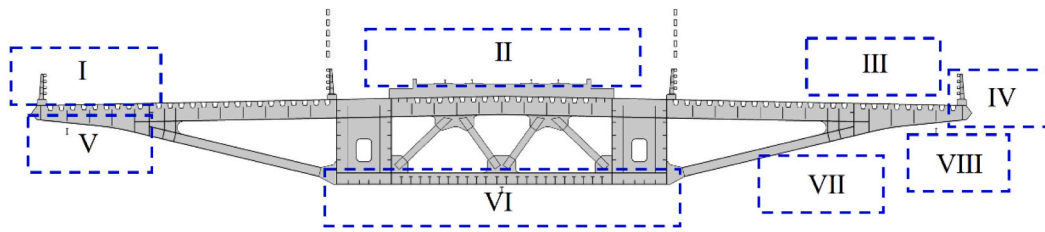
2.3. Wind tunnel set-up and test procedure

The experiments were conducted in the low-speed section of CSU-WT1 at Central South University, Changsha, China. The CSU-WT1 is a closed-circuit atmospheric boundary layer wind tunnel with two test

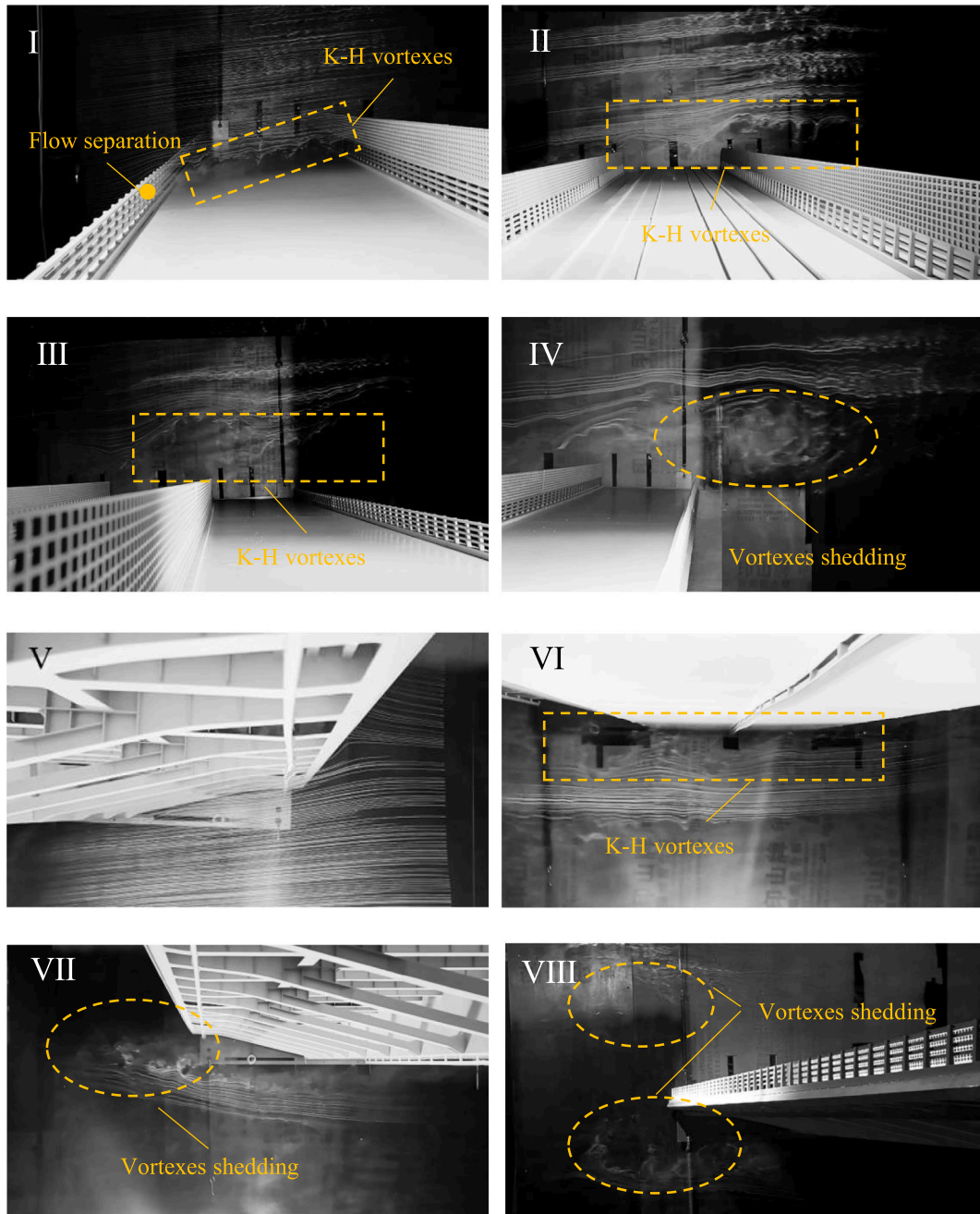
sections. At the location where the experiments were conducted, the low-speed section measured 12 m in width and 3.5 m in height, resulting in a maximum blocking ratio of 5.1 %. The wind speed ranged from 1 to 10 m/s with turbulence intensity less than 1 %. The corresponding Reynolds number was 1.046×10^4 – 1.046×10^5 , which was lower than that of the prototype. According to the study on the Great Belt East Bridge (Schewe et al. [49]), lower Strouhal numbers (St) were observed at low Reynolds number than at high Reynolds numbers. Consequently, VIV wind speeds measured in wind tunnel tests tend to be higher than those experienced by actual bridges. Furthermore, the turbulence intensity in the wind tunnel was also lower than that experienced by the prototype. Previous studies [50–56] have demonstrated that increasing Reynolds number and turbulence intensity generally reduce VIV amplitudes. Therefore, it is reasonable to conclude that wind tunnel test results tend to be conservative, biasing toward safety.

The bridge section model was suspended in the wind tunnel by steel arms connected to eight springs mounted in a support bracket, as shown in Fig. 2. Two wooden end plates are fixed to the support bracket at each end of the model to ensure a two-dimensional flow. The dynamic parameters were listed in Table 1. The model vertical and torsional frequencies in still air were 2.417 Hz and 4.699 Hz, respectively, with damping ratios of 0.27 % and 0.3 % for vertical and torsional vibrations, respectively.

The wind-induced vibrations of the section model were collected by three laser transducers, arranged underneath each side of the section model and aimed at reflexive targets set on the steel arms. The signals from the laser transducers were sampled at a rate of 1024 Hz for 30 s and then processed by a computer to extract the vertical and torsional



(a) Sketch of flow visualization positions



(b) Flow visualization

Fig. 9. Flow visualization for original bridge deck.

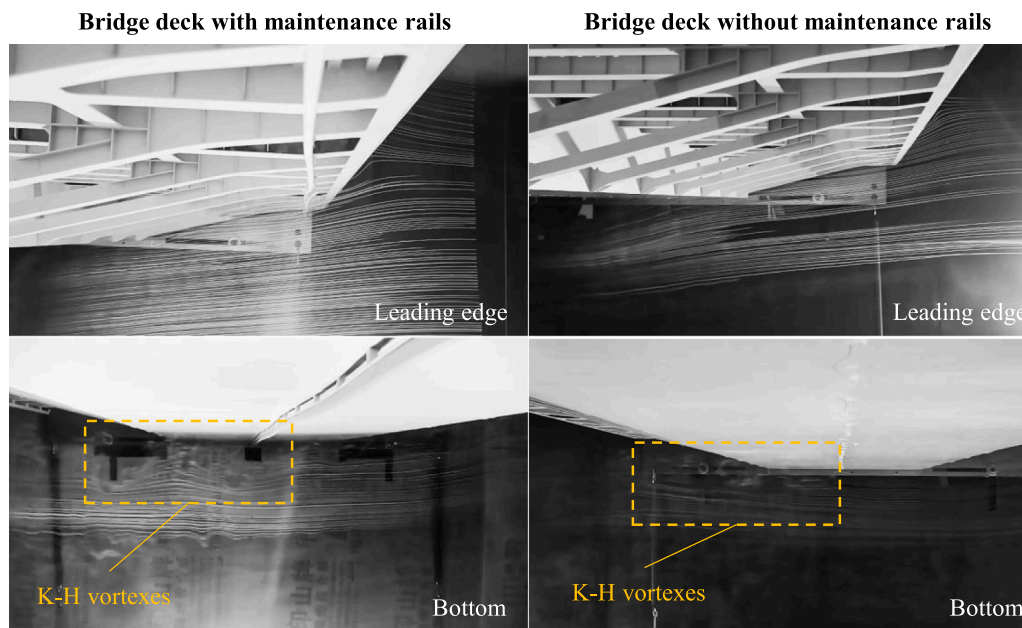


Fig. 10. Flow visualization for bridge deck with/without maintenance rails.

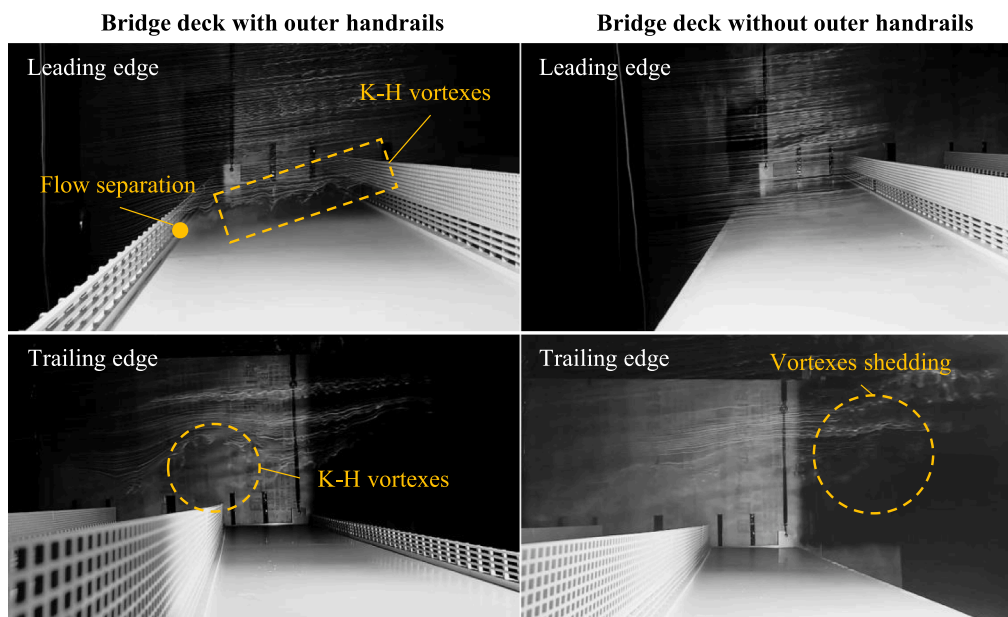


Fig. 11. Flow visualization for bridge deck with/without outer handrails.

vibrations. The free-stream wind speed was monitored 2 m in front of the section model using a TFI cobra probe with a sample rate of 1000 Hz.

To enhance the reliability of the test results, each test was repeated three times, and a fine scan of interest wind speed range was conducted with a wind speed interval of 0.5 m/s. In addition, during the wind speed of VIV range, the wind speed interval was further reduced. Vertical and torsional wind-induced vibrations were tested separately to ensure that the damping ratio was less than 0.3 % in each degree of freedom. The VIV performance of bridges is generally affected by the wind AoA, decreasing as the wind AoA increases [57–59]. Considering that the bridge is located in an open-sea environment, three wind angles of attack (AoAs) i.e., 0° and $\pm 3^\circ$, were selected to examine the VIV performance of the sectional model, in accordance with the recommendations of the Chinese specification [48]. The wind AoA were changed by adjusting the installation angle of the section model.

A smoke flow visualization was conducted to get a clear understanding of the flow pattern around the highway-railway same-level bridge girder. In addition, an experiment for the bridge deck without maintenance rails and outer handrails (Fig. 5.(a)) was performed to investigate the effects of the bridge appurtenances on the flow pattern and to aid the design of countermeasures.

3. VIV characteristics of original bridge deck

The evolution of the vertical and torsional amplitudes of the original bridge deck with increasing wind speed is shown in Fig. 3, where the error bars denote the standard deviation. According to the wind-resistant design specification [48], the permitted value of the vertical response is 0.140 m, while the permitted value of the torsional response is 0.167° . For the original bridge deck, apparent VIV ranges are observed

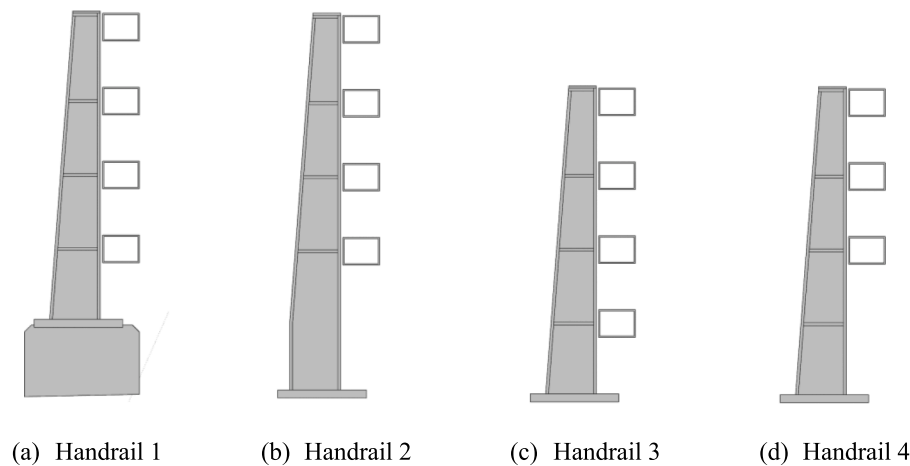


Fig. 12. Configurations of outer handrails.

Table 2

Description of outer handrails.

Case	Description
Handrail 1	Original handrails.
Handrail 2	Remove curbstones and keep the handrail height.
Handrail 3	Remove curbstones and reduce the handrail height.
Handrail 4	Remove curbstones and bottom railing and reduce the handrail height.

in both vertical and torsional vibrations. The wind speed range for vertical VIV is lower than that for torsional VIV, set at approximately 4 m/s, while the torsional VIV range is about 6.5 m/s. Notably, at a wind AoA of 0° , the VIV range is significant for both vertical and torsional vibrations, with amplitudes exceeding the permitted values given by the wind-resistant design specification [48]. At a wind AoA of 3° , the VIV ranges and amplitudes for vertical and torsional vibrations is decreased. The vertical VIV amplitude is slightly lower than the permitted value, while the torsional VIV amplitude is still higher than the permitted value. At a wind AoA of -3° , although both VIV ranges are still observed, the range width and amplitude are further reduced. Both the vertical and torsional VIV amplitudes are lower than the permitted values.

The spectra as a function of wind speed for the both vertical and torsional vibrations are shown in Fig. 4. Predominant frequency ranges are observed in both vertical and torsional vibrations, which are equal to vertical and torsional frequencies, respectively. The predominant frequency ranges show a high consistency with the VIV ranges. This indicates that the two VIV ranges are completely separated, with vertical bending and torsional modes dominating, respectively.

4. Investigation on causes of VIV

4.1. Effect of maintenance rails and outer handrails on VIV performance

The previous studies [30,33,34,45] showed that the VIV performance of box girder bridge decks was significantly affected by maintenance rails and outer handrails. This is because the fact that maintenance rails and outer handrails influence the leading and trailing flow separations, thereby changing the vortex shedding pattern and controlling the VIV performance. Given this, the bridge deck without maintenance rails and outer handrails (see Fig. 5.(a)) was first tested to assess their influence on the VIV performance of the single-box girder with long cantilevers. The corresponding vibration amplitudes are presented in Fig. 6. It is obvious that the vertical and torsional amplitudes are lower than the permitted values at the tested wind AoAs. The vertical VIV disappears completely, as no significant VIV ranges are observed

within the tested wind speeds. The torsional VIV still occurs at the wind AoAs of $\pm 3^\circ$, but the amplitudes are considerably lower than that of the original bridge deck and are almost equivalent to the slight random motions. The results indicate that maintenance rails and handrails have a prominent effect on the VIV performance of the bridge deck, and have the potential to control the VIV by modifying their arrangements.

4.2. Effects of maintenance rails on VIV performance

The previous section indicates that the maintenance rails and the outer handrails are the primary contributors to the large amplitude VIV observed in the original bridge deck. However, the exact component responsible for this effect remains unidentified. In this section, the maintenance rails were progressively removed to investigate their effect on the VIV performance. The bottom maintenance rails were removed first, followed by the maintenance rails on both sides, as shown in Fig. 5. (b)-(c). Given that the VIV amplitudes of the original bridge deck are most significant at the wind AoA of 0° compared to other tested wind AoAs, the experiments were conducted specifically at the wind AoA of 0° .

The amplitudes of the vertical and torsional vibrations of bridge deck without maintenance rails are shown in Fig. 7. After the removal of bottom maintenance rails, the torsional amplitudes are almost the same as for the original bridge deck, with a torsion peak of 0.560° . However, the vertical amplitudes exhibit a little difference. Removing the bottom maintenance rails leads to a slight reduction in the bending peak, decreasing from 0.149 m to 0.118 m, which falls below the permitted value. For the removal of all maintenance rails, the bending peak is similar to that of the original bridge deck. While the torsion peak shows a slight decrease from 0.546° to 0.434° , its value remains higher than the permitted value.

The removal of maintenance rails has limited effects on the VIV performance. Although the removal of all maintenance rails decreases the torsion peak, the amplitude value remains significantly higher than the permitted value. Additionally, the presence of the maintenance rails may have a positive effect on the VIV performance, as evidenced by the higher bending peak observed in the bridge deck without all maintenance rails compared to the bridge deck without bottom maintenance rails. Therefore, although the maintenance rails influence the VIV performance, they are not the primary cause of VIV in the original bridge deck.

4.3. Effect of outer handrails on VIV performance

To investigate the influence of handrails on the VIV performance, the outer handrails were removed while retaining the inner handrails,

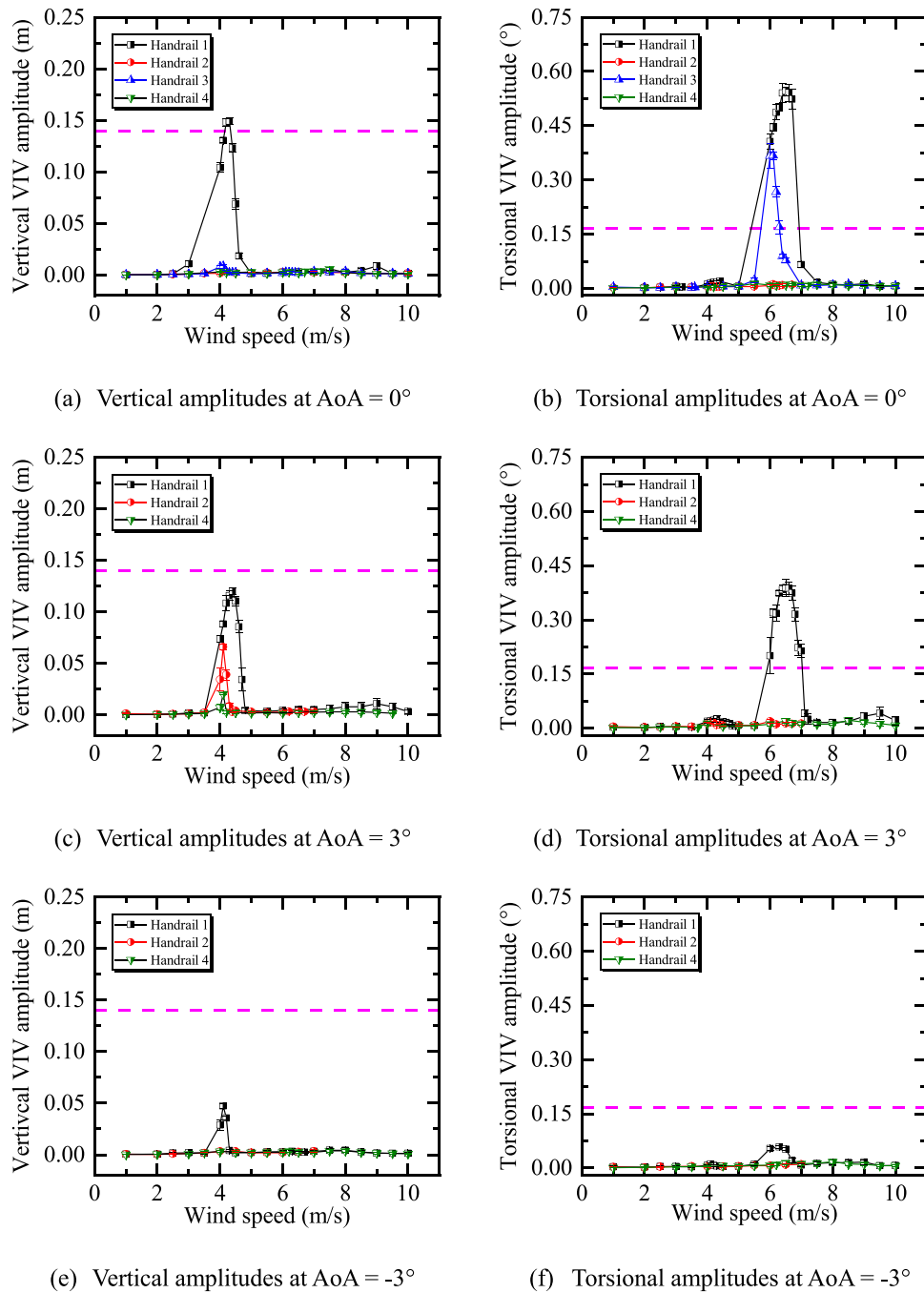


Fig. 13. VIV amplitudes for bridge deck with various outer handrails.

lighter barriers and all maintenance rails, as shown in Fig. 5.(d). The experiments were conducted at the wind AoA of 0° , the same condition as in the previous section.

The evolution of the amplitudes as a function of wind speed for the bridge deck without outer handrails is shown in Fig. 8. The removal of the outer handrails has a significant effect on controlling the VIV of the bridge deck. Both vertical and torsional VIVs completely vanish, as no obvious amplitude is observed within the tested wind speeds.

The results suggest that the outer handrails are the main cause for the VIV of the original bridge deck. Therefore, modifying the outer handrails could be considered an effective approach to controlling the VIV of the original bridge deck.

4.4. Smoke flow visualization

Smoke flow visualization is an essential tool for understanding flow behavior. While smoke flow visualization may not provide precise data on the flow field, it offers a qualitative macroscopic description of flow pattern. A clear understanding of the flow pattern around the bridge girder is helpful in effectively developing aerodynamic countermeasures.

Fig. 9 shows a smoke flow visualization for the original bridge deck at the wind AoA of 0° . On the upper surface, the flow separates at the outer handrail on the leading edge and reattaches to the surface, generating Kelvin-Helmholtz (K-H) vortices. As these vortices move downstream and eventually separate from the trailing edge, vortex shedding develops in the wake. Similarly, on the lower surface, the flow

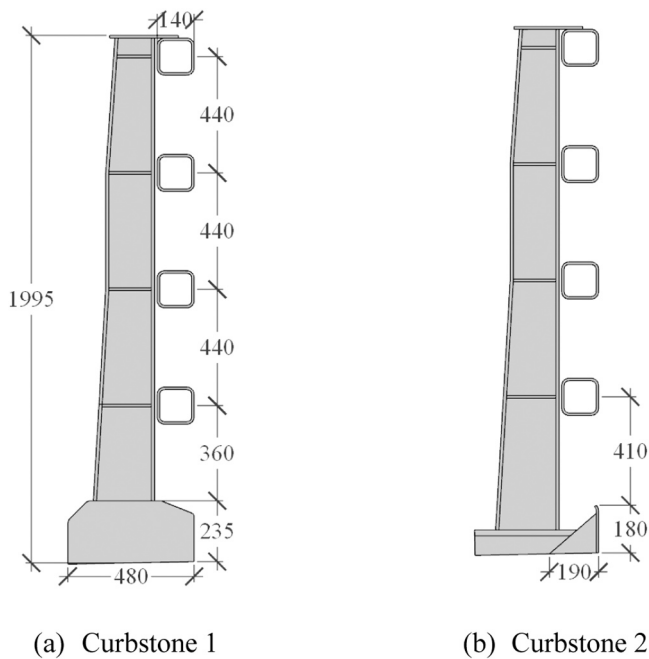


Fig. 14. Outer handrail with new curbstones (unit: mm).

separates at the windward corner of the bottom box and generates K-H vortices, leading to vortex shedding at the trailing edge. As alternating vortex shedding from the upper and lower surfaces, the Von Kármán vortex is formed in the wake, inducing the occurrence of VIV.

Fig. 10 show the flow field for the bridge deck with/without the maintenance rails. From Fig. 10 it is noted that the existence of the outer maintenance rails has a little effect on the flow field around the leading edge. The flow passes smoothly over the leading edge, exhibiting consistent behavior regardless of whether the outer maintenance rails are present or absent. As for the flow field around the bottom of the bridge deck, the occurrence of the K-H vortices is primarily attributed to the bottom box of the bridge deck. The K-H vortices at the bottom of the bridge deck are observed in both cases.

The flow field around the bridge deck with and without the outer handrails is completely different, as shown in Fig. 11. In the original bridge deck, K-H vortices are generated from the curbstone of the outer handrails. Whereas in the bridge deck without outer handrails, the flow passes smoothly over the leading edge. Although K-H vortices are observed on the trailing edge in both cases, their scale is noticeably larger for the original deck. For the bridge deck without outer handrails, the observed K-H vortices are likely induced by the inner handrails and light barriers, which shift the formation of large coherent vortices associated with Von Kármán vortex farther downstream. As a result, the bridge deck is less susceptible to direct influence from vortex shedding.

The smoke flow visualization as shown in Figs. 9–11 indicates that the VIV is mainly caused by the vortices induced by the outer handrails, and the curbstone of the outer handrails plays an important role in vortex formation. Therefore, the design of the aerodynamic countermeasures should focus on the arrangement of the outer handrail, especially the aerodynamic configuration of the curbstone.

5. Aerodynamic profile optimization of outer handrails

5.1. Effect of curbstone

As discussed in the previous section, the outer handrails have a significant effect on the VIV performance of the bridge deck, and the smoke flow visualization further points out that the curbstone at the bottom of outer handrails may have significant effects on vortex generation.

Hence, a series of further wind tunnel experiments were conducted to study the effect of the curbstone. Four configurations of the outer handrails were singled out for testing, as shown in Fig. 12 and summarized in Table 2. Handrail 1 represents the original configuration installed on the bridge deck. In contrast, Handrails 2, 3, and 4, were modified by removing the curbstone at the bottom of the outer handrails. Among these, Handrail 2 maintained the same height as Handrail 1, while the height of Handrail 3 decreased with the removal of the curbs. Handrail 4 shared the same height as Handrail 3; however, its bottom railing was also removed to increase the porosity ratio at the bottom of the handrails, further altering its aerodynamic characteristics. It should be noted that only the outer handrails were replaced in these experiments, and other bridge appurtenances were consistent with the original bridge deck.

The VIV performance was first evaluated at the wind AoA of 0° , where the VIV amplitudes of the original bridge deck were the largest. The amplitudes as a function of wind speed are shown in Fig. 13.(a)-(b). By comparison to Handrail 1, it is noted that the torsional VIV still occurred in Handrail 3 with the curbstone removed. While the torsion peak decreases from 0.546° to 0.367° , it still exceeds the permitted value. However, for the bridge deck with Handrail 2 and 4, no obvious VIV amplitudes are observed in both vertical bending and torsion. These results are consistent with the observations from the smoke flow visualization, and provide further evidence that the curbstone is a crucial factor in the VIV performance of the bridge deck. Furthermore, the occurrence of VIV seems to be primarily related to the porosity at the bottom of the handrails. This inference is supported by the observation that VIVs vanish in configurations featuring a high bottom porosity (Handrail 2 and 4), while VIVs resurface in Handrail 3, characterized by a low bottom porosity. Therefore, the optimization of the outer handrails involves not only the design of the curbstone but also the consideration of the porosity ratio at the bottom of the handrails.

As Handrail 3 demonstrates an obvious VIV in torsion, further experiments were specifically carried out for Handrails 2 and 4. The VIV performance of Handrails 2 and 4 was evaluated at the wind AoAs of $\pm 3^\circ$, and Fig. 13.(c)-(f) show the results of these experiments. From Fig. 13.(c), bending peaks at roughly the wind speed of 4 m/s is observed for both Handrails 2 and 4, with the values much less than the permitted value. Handrails 2 and 4 have the effects of significantly reducing the bending peak. Specifically, Handrail 2 reduces the bending peak from 0.120 m to 0.066 m, while Handrail 4 reduces the bending peak from 0.120 m to 0.020 m. As for the other cases, no obvious amplitude of the VIV is observed. The results indicate that the curbstone has a significant effect on the VIV performance, not only for the wind AoA of 0° but also for wind AoAs of $\pm 3^\circ$.

While removing the curbstone and increasing the bottom porosity of the outer handrails can effectively mitigate the VIV of the bridge deck, the curbstone plays a crucial role as a drainage facility. Hence, there is still a requirement for optimizing the aerodynamic configurations of the outer handrails to improve the VIV performance.

5.2. Handrail optimization

In the previous section, it was found that the curbstone aerodynamic shape and the porosity ratio at the bottom of the handrails were mainly responsible for the VIV of the bridge deck. To mitigate the VIV, the outer handrails had been modified. The redesigned outer handrail featured a higher porosity ratio compared to the original one, as shown in Fig. 14. Railing spacing was expanded from 320 mm to 440 mm, and the handrail height was raised from 1665 mm to 1995 mm. In addition, two new curbstones were designed to improve the bottom porosity and the aerodynamic profile. By comparison to the original curbstone, the Curbstone 1 streamlined the upper corners and reduced the height from 305 mm to 235 mm. In the Curbstone 2 configuration, the curbstone was replaced by a baffle plate to maximize the bottom porosity of the outer handrails.

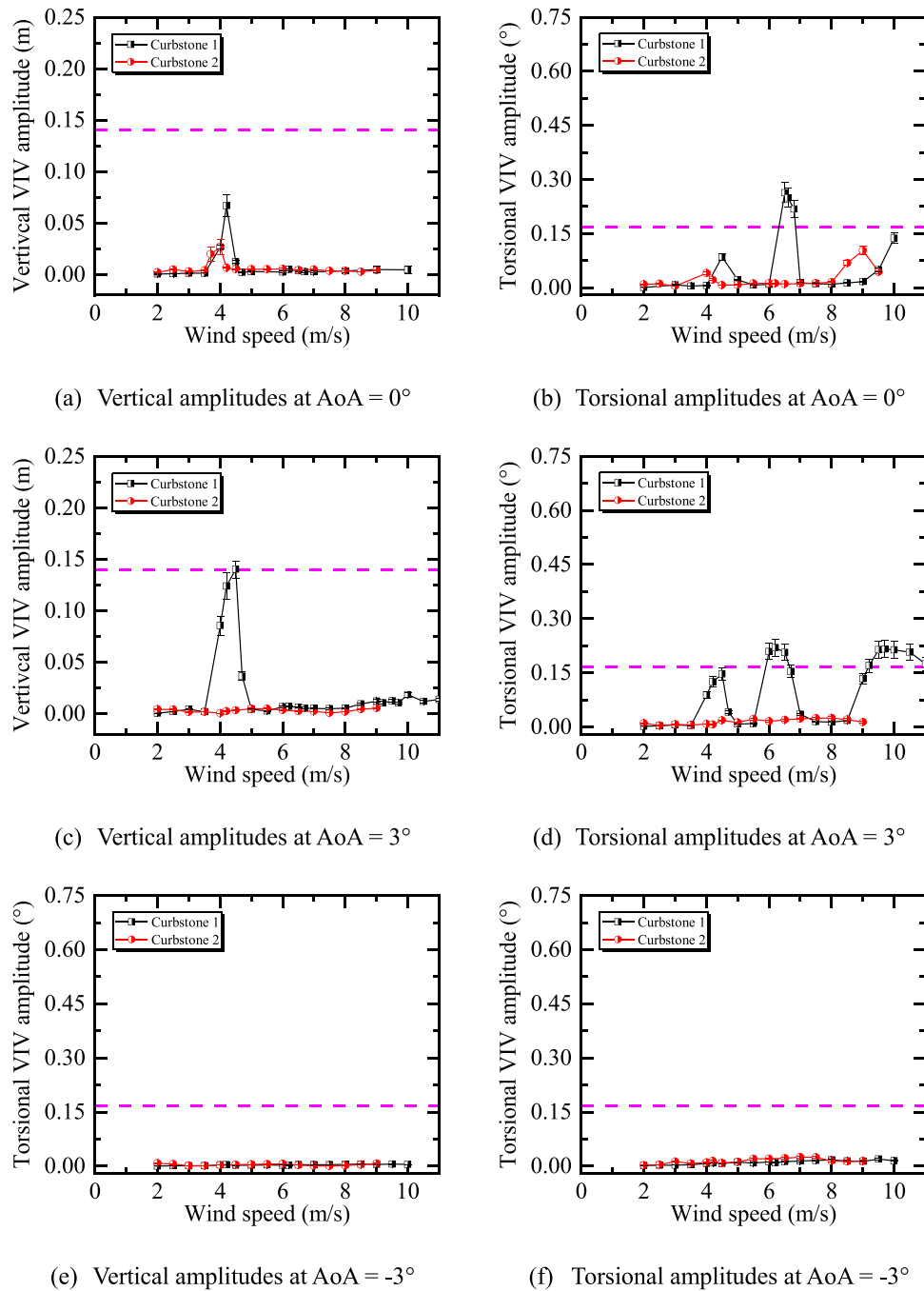


Fig. 15. VIV amplitudes for outer handrail with new curbstones.

A series of wind tunnel experiments for the bridge deck installed with new outer handrails were conducted to evaluate the VIV performance. Fig. 15 shows the evolution of vertical and torsional amplitudes with varying wind speeds. The VIV amplitudes are completely vanished at the wind AoA of -3° . This indicates that the configurations of Curbstone 1 and 2 successfully mitigate the VIVs of the bridge deck. As for other wind AoAs, the results differ for configurations. For the curbstone 1 configuration, while the amplitudes are decreased a lot in both vertical bending and torsion, the torsional amplitudes surpass the permitted value at the wind AoAs of 0° and 3° . Moreover, the bending peak is equal to the permitted value at the wind AoA of 3° . For the Curbstone 2 configuration, despite the occurrence of VIVs in both vertical bending and torsion at the wind AoA of 0° , the VIV amplitudes are far below the permitted value, with a bending peak of 0.026 m and a torsion peak of

0.103° . Therefore, the Curbstone 2 configuration was selected as the aerodynamic countermeasure.

6. Conclusions

In the present study, the VIV performance of a highway–railway same-level bridge, characterized by a single-box girder with extended cantilevers, was experimentally investigated. Utilizing flow visualization techniques, the primary factors affecting VIV performance were successfully identified. The findings demonstrate that modifications to the handrail configuration can effectively mitigate the VIVs in this bridge type. The main conclusions are summarized as follows:

- (1) The bridge is prone to VIVs in both vertical bending and torsion. The VIV ranges in both degrees of freedom are separate, and the VIV peaks surpass the permitted values.
- (2) The VIV performance of the bridge deck is influenced by both the outer handrails and maintenance rails, with the outer handrails playing a dominant role. Smoke flow visualization provides significant observations, indicating that the large amplitude VIVs are highly associated with the curbstone and the bottom porosity of the outer handrails.
- (3) The modification of the outer handrails can effectively mitigate the VIVs of the bridge deck. The curbstone aerodynamic shape and the bottom porosity of the outer handrails are two decisive factors in modifying the outer handrails. Streamlining the curbstone aerodynamic shape and improving the bottom porosity ratio can significantly reduce the VIV amplitudes in both vertical bending and torsion.

CRedit authorship contribution statement

Lingfei Tan: Formal analysis, Data curation. **Haiquan Jing:** Writing – review & editing, Methodology, Investigation, Conceptualization. **Naeem Muhammad:** Writing – review & editing. **Xuhui He:** Supervision, Resources, Project administration. **Zhen Wang:** Writing – original draft, Validation, Software, Methodology. **Lei Yan:** Writing – review & editing, Methodology, Conceptualization. **Lulu Liu:** Validation, Investigation.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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